

Data Collection and Preprocessing Phase

Date	16 August 2026
Team ID	XXXXXX
Project Title	India's Agricultural Crop Production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	• Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. • Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.
Data Cleaning	• Null Value Handling: Imputed missing values in Production using regional median yield rates; dropped records missing essential spatial identifiers. • Outlier & Noise Reduction: Removed corrupt entries with negative area values or impossible yield spikes caused by data entry errors
Data Transformation	Calculated the standardized productivity metric $\text{Yield (Tones/Hectare)} = \frac{\text{Production}}{\text{Area}}.$ Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.

Data Type Conversion	• Numerical Casts: Explicitly converted Area and Production to floating-point numeric types to allow statistical aggregations. • Temporal & Categorical Casting: Formatted Crop Year into standard integer date format and converted categorical text fields into strings for dynamic visualization filtering.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).
Data Modeling	☐ Star Schema Architecture: Created a central Fact Tab linked to look up Dimension Tables. ☐ Relationship Building: Established primary-foreign key relationships (1: N) between dimensions and facts to optimize query speed and drill-down performance in Tableau.
Save Processed Data	☐ Storage Format: Exported clean data to optimized .csv /.parquet formats and pushed structured tables to SQL database staging tables. ☐ Tableau Extraction: Created a hyper-extracted data source file (.hyper) ready for rapid visualization render in